Haoyuan Li

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EDUCATION

University of North Carolina at Chapel Hill, Chapel Hill, US Sep.2022-May 2027(expected)

Major: Computer Science, Doctor of Philosophy, Department of Computer Science

- **Advisor: Snigdha Chaturvedi**

Harvard University, Cambridge, US Sep.2020-Jan.2022

Major: Health Data Science, Master of Science, School of Public Health

- **Cumulative GPA:** 3.79/4.0

Renmin University of China, Beijing, CN Sep.2016–July.2020

Major: Statistics, Bachelor of Science, School of Statistics

- **Cumulative GPA:** 3.74/4.0

INTERNSHIP

Amazon May.2026-Aug.2026

- **Position:** Applied Scientist Intern **Mentor:** Shuai Wang

Amazon May.2025-Aug.2025

- **Position:** Applied Scientist Intern **Mentor:** Zhengyuan Shen

Microsoft Research Asia Jan.2021-July.2021

- **Position:** Research Intern **Mentor:** He Di

Yale University Jun.2020-Sep.2020

- **Position:** Research Assistant **Advisor:** Dragomir R. Radev

Stanford University Jun.2019-Sep.2019

- **Position:** Research Assistant **Advisor:** Ram Rajagopal

PUBLICATIONS

Haoyuan Li, Snigdha Chaturvedi, "Inter-dimension Dependence for Multi-Dimensional Evaluation of Open-Ended Text", submitted to EACL 2027

Haoyuan Li, Zhengyuan Shen, Sullam Jeoung, Yueyan Chen, Jiayu Li, Qi Zhu, Shuai Wang, Vassilis N. Ioannidis, Huzefa Rangwala, "BoundRL: Efficient Structured Text Segmentation through Reinforced Boundary Generation", ACL 2026 Findings

Haoyuan Li, Snigdha Chaturvedi, "Comparative Personalization for Multi-document Summarization", Submitted to AACL 2026

Haoyuan Li, Rui Zhang, Snigdha Chaturvedi, 'Improving Fairness of Large Language Models in Multi-document Summarization', ACL 2025

Haoyuan Li, Yusen Zhang, Rui Zhang, Snigdha Chaturvedi, 'Coverage-based Fairness in Multi-document Summarization', NAACL 2025 SAC award

Haoyuan Li, Snigdha Chaturvedi, 'Rationale-based Opinion Summarization', NAACL 2024

Haoyuan Li, Somnath Basu Roy Chowdhury, Snigdha Chaturvedi, 'Aspect-aware Unsupervised Extractive Opinion Summarization', ACL 2023 Findings

Bi, Qiwei, **Haoyuan Li**, and Hanfang Yang. 2021. "Boosting Few-Shot Abstractive Summarization with Auxiliary Tasks.", CIKM 2021

Fabbri, Alexander R., Simeng Han, **Haoyuan Li**, Haoran Li, Marjan Ghazvininejad, Shafiq Joty, Dragomir Radev, and Yashar Mehdad. "Improving Zero and Few-Shot Abstractive Summarization with Intermediate Fine-tuning and Data Augmentation.", NAACL 2021

Wang, Z.*, **Haoyuan Li***, Rajagopal, R., (2019) Urban2Vec: Incorporating Street View Imagery and POIs for Multi-Modal Urban Neighborhood Embedding, AAAI 2020, (*contribute equally)

REASEARCH & PROJECTS

Inter-dimension Dependence for Multi-Dimensional Evaluation of Open-Ended Text

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: We propose **CorrGap**, a metric that quantifies interference from unrelated evaluation dimensions in LLM-as-a-Judge frameworks for open-ended text. We show that inter-dimension interference is pervasive across multiple generation tasks and evaluation frameworks, revealing a fundamental limitation of multi-dimensional LLM evaluation. To improve judge reliability, we propose **DimCheck**, which removes evidence unrelated to the target dimension from the judge's chain-of-thought reasoning.

BoundRL: Efficient Structured Text Segmentation through Reinforced Boundary Generation

Advisor: Shen Zhengyuan, Amazon

Summary: We propose **BoundRL**, an efficient framework for token-level segmentation and label prediction of long and complex LLM prompts. Rather than generating the full content of each segment, BoundRL predicts only segment boundaries and starting tokens, substantially reducing generation overhead. We adapt the model with **reinforcement learning with verifiable rewards (RLVR)**, jointly optimizing document reconstruction fidelity and semantic alignment. To mitigate entropy collapse, we introduce intermediate candidates through systematic perturbation of generated segment sequences, providing stepping stones toward higher-quality solutions.

Comparative Personalization for Multi-document Summarization

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: We propose **ComPSum**, a framework for personalized multi-document summarization that models user preferences through comparative analysis. The system first constructs a structured user preference profile by contrasting the target user's preferences with those of other users, and then conditions summary generation on the resulting profile. We further propose **AuthorMap**, a fine-grained, reference-free evaluation framework that measures personalization through authorship attribution between summaries generated for different users.

Improving Fairness of Large Language Models in Multi-document Summarization

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: We propose **FairPO**, a preference optimization framework for improving fairness in LLM-generated multi-document summaries. We construct preference pairs by generating summaries from perturbed document sets and train the model using a DPO-based objective. To improve corpus-level fairness, we dynamically reweight preference pairs according to their contribution to the target fairness objective.

Coverage-based Fairness in Multi-Document Summarization

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: We introduce **Equal Coverage**, a summary-level fairness metric that evaluates whether documents associated with different social attributes receive equitable coverage in multi-document summaries. Unlike prior measures such as Proportional Representation, Equal Coverage accounts for redundancy among selected content. We further propose **Coverage Parity**, a corpus-level metric for measuring fairness across a collection of summaries.

Rationale-based Opinion Summarization

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: We introduce Rationale-based Opinion Summarization, a framework that pairs each representative opinion with a supporting rationale to improve the usefulness and interpretability of opinion summaries. We identify rationales using four complementary criteria: relatedness, specificity, popularity, and diversity.

Aspect-aware Unsupervised Extractive Opinion Summarization

Advisor: Snigdha Chaturvedi, University of North Carolina at Chapel Hill

Summary: We propose an unsupervised extractive opinion summarization method that explicitly models **aspect diversity** to improve coverage of heterogeneous opinions. The method automatically discovers latent aspects from noun phrases and adjectives in review sentences and selects representative sentences across different aspects to construct the summary.

Boosting few-shot abstractive summarization with auxiliary tasks

Advisor: Hanfang Yang, Renmin University of China

Summary: We improve few-shot abstractive summarization through multi-task learning with three auxiliary objectives: **sentence extraction, object prediction, and triple entailment**. I developed a BART-based summarization model with adapters for multi-task learning, conducted experiments on **XSum**, and contributed to the paper.

Improving Zero and Few-Shot Abstractive Summarization with Intermediate Fine-tuning and Data Augmentation

Advisor: Dragomir R. Radev, Yale University

Summary: We develop a domain-adaptation framework for zero- and few-shot abstractive summarization by generating pseudo-training pairs from Wikipedia that match the target dataset's characteristics, such as abstractiveness. We further incorporate **back-translation and consistency regularization** to improve robustness to limited supervision. I designed the consistency-loss objective between original and augmented documents and contributed to the paper.

Urban2Vec: Incorporating Street View Imagery and POIs for Multi-Modal Urban Neighborhood Embedding

Advisor: Ram Rajagopal, Stanford University

Summary: We develop **Urban2Vec**, an unsupervised multimodal representation learning framework that combines **street-view imagery and points of interest (POIs)** to encode urban neighborhoods. We evaluate the learned representations through downstream demographic prediction and neighborhood clustering. I designed the image-text fusion approach and developed evaluation methods for the learned representations.